

20 Jan 2026

$$\dot{x} = f(x)$$

Definition: $x(t)$ is a periodic orbit if $\exists T > 0$ s.t. $x(t+T) = x(t) \forall t$

Linear Systems:

1D

$$\dot{x} = a x$$

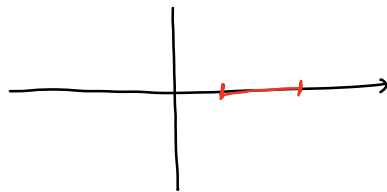
No periodic orbits $\forall a$

Non-linear Systems:

1D

$$\dot{x} = f(x)$$

No periodic orbits



$f(x)$ along $x(t)$ is monotonic

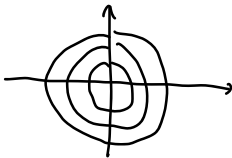
Linear Systems:

2D

$$\dot{x} = \begin{bmatrix} 0 & -\beta \\ \beta & 0 \end{bmatrix} x$$

$$\lambda_{1,2} = \pm j\beta$$

$$\begin{cases} \dot{x}_1 = -\beta x_2 & \ddot{x}_1 = -\beta \dot{x}_2 \\ \dot{x}_2 = \beta x_1 & \ddot{x}_2 = \beta \dot{x}_1 \end{cases}$$



$$\frac{d}{dt} (x_1^2 + x_2^2) = 2x_1 \dot{x}_1 + 2x_2 \dot{x}_2$$

$$= 2x_1(-\beta x_2) + 2x_2(\beta x_1)$$

$$= 0$$

an "isolated" periodic orbit cannot occur in linear system

limit cycle = "isolated" periodic orbit

$$\dot{x} = Ax$$

$$\underline{20} \quad z = T^{-1}x \Rightarrow \dot{z} = T^{-1}\dot{x} = T^{-1}Ax = T^{-1}ATz$$

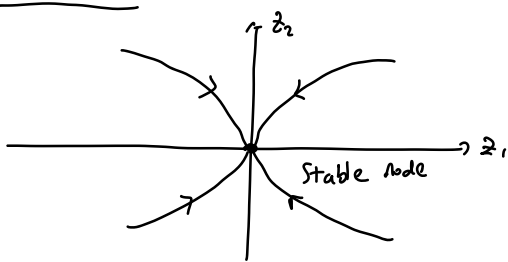
for proper choice of T

$$\dot{z} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} z$$

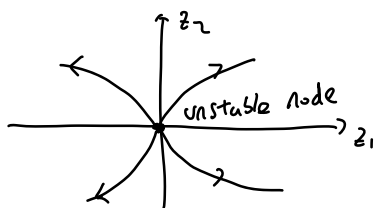
$$\text{or } \dot{z} = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} z$$

$$\text{or } \dot{z} = \underbrace{\begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}}_{T^{-1}AT} z$$

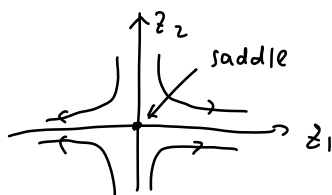
Real λ 's



$$\lambda_1 < \lambda_2 < 0$$



$$\lambda_1 > \lambda_2 > 0$$



$$\lambda_2 < 0 < \lambda_1$$

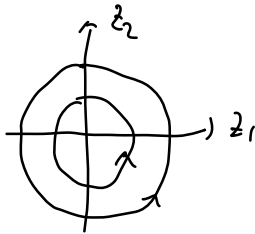
$$\dot{z} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} z$$

$$\begin{aligned} \dot{z}_1 &= \lambda_1 z_1 & \Rightarrow & z_1 = e^{\lambda_1 t} z_1(0) \\ \dot{z}_2 &= \lambda_2 z_2 & \Rightarrow & z_2 = e^{\lambda_2 t} z_2(0) \end{aligned}$$

Complex eigenvalues

$$\dot{z} = \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix} z$$

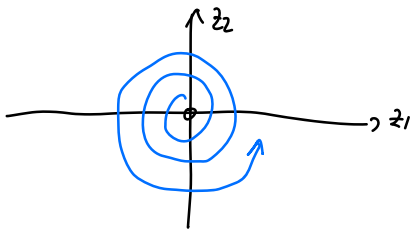
$$\alpha = 0$$



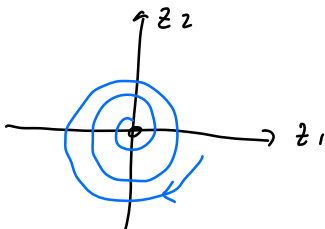
"marginally stable"

$$\alpha > 0$$

$$\begin{aligned} \frac{d}{dt} (z_1^2 + z_2^2) &= 2z_1 \dot{z}_1 + 2z_2 \dot{z}_2 \\ &= 2z_1 (\alpha z_1 - \beta z_2) + 2z_2 (\beta z_1 + \alpha z_2) \\ &= 2\alpha (z_1^2 + z_2^2) \end{aligned}$$



$$\alpha < 0$$



$$\dot{x} = Ax, \quad A \in \mathbb{R}^{n \times n}$$

$$\operatorname{Re}(\lambda_i) < 0 \quad \forall i$$

$\Rightarrow$ stable eq.

$$\operatorname{Re}(\lambda_i) > 0 \quad \text{for at least one } i$$

$\Rightarrow$ unstable eq.

$$\operatorname{Re}(\lambda_i) \leq 0 \quad \forall i \quad \text{and no repeated}$$

$\Rightarrow$ "marginally stable"

e-values on the $j\omega$ axis

Ex. $\dot{z} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \quad \lambda_{1,2} = 0 \quad \text{"unstable system"}$

$$\dot{z}_1 = z_2 \quad z_1(t) = z_2(0)t + z_1(0)$$

$$\dot{z}_2 = 0 \quad z_2(t) = z_2(0)$$

Hartman-Grobman Theorem

Provides basis for qualitative comparison between $\dot{x} = f(x)$ and $\dot{x} = \frac{\partial f}{\partial x} \Big|_{x=x^*} (x - x^*)$

Thm: If x^* is a hyperbolic eq

no e-value on the imaginary axis

Then there exists a **homeomorphism**

$z = h(x)$ defined in a neighborhood of x^* that maps trajectories of $\dot{x} = f(x)$

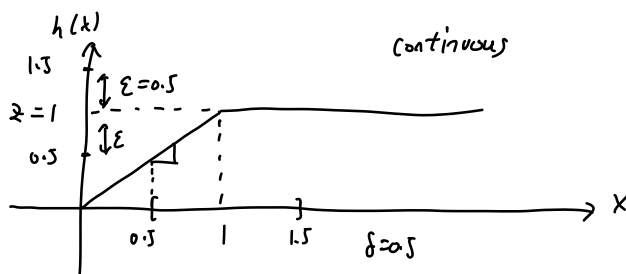
to those of $\dot{z} = Az$. $A = \frac{\partial f}{\partial x} \Big|_{x=x^*}$

homeomorphism: a continuous map with a continuous inverse

$z = h(x)$ is continuous at x if $\forall \epsilon > 0$

$\exists \delta > 0$ s.t. $\| \tilde{x} - x \| < \delta \Rightarrow \| \tilde{z} - z \| < \epsilon$

$$\tilde{z} = h(\tilde{x})$$

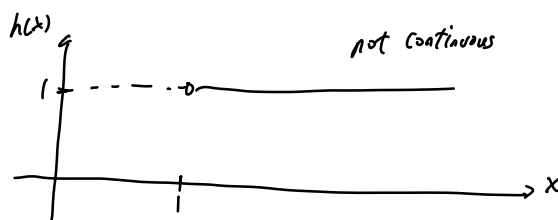


$$x=1, z=1$$

$$\epsilon = 0.5$$

$$\delta = 0.5$$

$$\| \tilde{x} - x \| \leq 0.5 \rightarrow \| \tilde{z} - z \| \leq 0.5$$

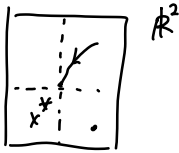


$\epsilon = 0.5$ (user specified)

pick δ

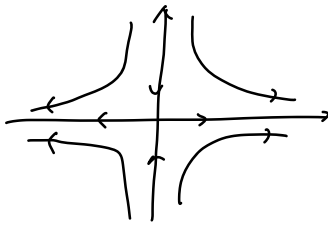
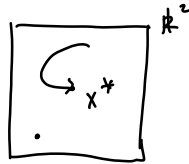
$$\| \tilde{x} - 1 \| < \delta \Rightarrow \| \tilde{z} - 1 \| < 0.5$$

$$\dot{\tilde{z}} = A\tilde{z}$$



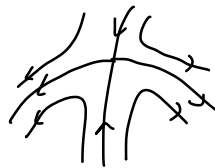
$$\tilde{z} = h(k)$$

$$\dot{x} = f(x)$$



Saddle point

$\Rightarrow$



Ex. necessity of hyperbolicity condition

$$\dot{x}_1 = -x_2 + a x_1 (x_1^2 + x_2^2)$$

$$\dot{x}_2 = x_1 + a x_2 (x_1^2 + x_2^2)$$

$$x_1 = x_2 = 0$$

$$\frac{\partial f_1}{\partial x_1} = 3a x_1^2 \quad \Big|_{x=x^*} = 0$$

$$\frac{\partial f_1}{\partial x_2} = 2a x_1 x_2 - 1 \quad \Big|_{x=x^*} = -1$$

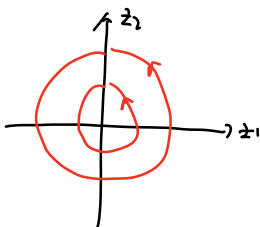
$$\frac{\partial f_2}{\partial x_1} = 2a x_1 x_2 + 1 \quad \Big|_{x=x^*} = 1$$

$$\frac{\partial f_2}{\partial x_2} = 3a x_2^2 \quad \Big|_{x=x^*} = 0$$

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \lambda^2 + 1 = 0$$

$$\lambda = \pm j$$

$$\dot{z} = Az$$



$z(nt) - z(1)$ is very small

$$x(nt) - x(z)$$

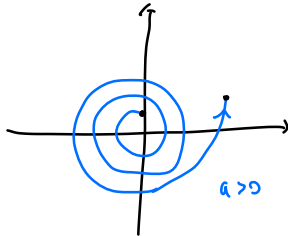
$$\dot{x} = f(x)$$

$$r = x_1^2 + x_2^2$$

$$\theta = \tan^{-1} \left(\frac{x_2}{x_1} \right)$$

$$\dot{r} = ar^3$$

$$\dot{\theta} = 1$$



$$\dot{x} = f(x)$$



$$\dot{z} = Az$$

$$A = \frac{\partial f}{\partial x} \Big|_{x=\bar{x}=0}$$

- $\bar{x}=0$ is (locally)

asymptotically stable

$$\operatorname{Re}\{k_i\} < 0 \quad \forall i$$

- $\bar{x}=0$ is unstable



$$\operatorname{Re}\{k_i\} > 0 \text{ for some } i$$

$\exists x_0$ st. $x(t)$ grows unbounded

End of Linearization

1D finite escape time

$$\dot{x} = x^2 \quad x=0 \text{ eq.}$$

$$\int_{x_0}^x \frac{dx}{x^2} = \int_{t_0}^t dt$$

$$-\frac{1}{x} \Big|_{x_0}^x = t - t_0$$

$$\frac{1}{x_0} - \frac{1}{x} = t - t_0$$

$$\Rightarrow \frac{1}{x} = \frac{1}{x_0} - (t - t_0)$$

$$\frac{1}{x} = \frac{1 - x_0(t - t_0)}{x_0}$$

$$x = \frac{x_0}{1 - x_0(t - t_0)}$$

$$t_0 = 0, \quad x(t) = \frac{1}{\frac{1}{x_0} - t}$$

$$t \rightarrow \frac{1}{x_0}$$

$$\Rightarrow x(t) \rightarrow \infty$$

1D $\dot{y} = y^2 + 1 \quad y \in \mathbb{R}$

No eq.